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Experiment 9

Aim: Install Docker and execute docker commands to manage images and interact with containers.

Theory:

What is Docker?

Docker is a platform that allows developers to build, package, and deploy applications in lightweight, portable containers. These containers include everything needed to run an application, such as code, runtime, system tools, libraries, and dependencies.

Containerization Technology

Containers are isolated environments where applications run independently. Unlike traditional virtualization, which requires separate operating systems for each application, containers share the host OS kernel, making them faster and more efficient.

Docker Architecture

Docker follows a client-server architecture consisting of the following key components:

1. Docker Client
   - It is the command-line interface (CLI) that allows users to interact with Docker.
   - Commands such as docker run, docker build, and docker stop are executed through the client.
2. Docker Daemon (dockerd)
   - It runs in the background and manages Docker containers, images, volumes, and networks.
   - It listens for requests from the Docker Client and executes commands.

3. Docker Images
    - A Docker image is a read-only template containing the application code, libraries, and dependencies.
    - Images are created using Dockerfiles, which define the steps to build an image.
    - Images are stored in Docker Hub or private repositories.
4. Docker Containers
    - A container is an instance of a Docker image running as an isolated process on a host machine.
    - Containers are lightweight, portable, and can be started, stopped, or removed as needed.
5. Docker Registry
    - It is a storage system for Docker images.
    - The public registry, Docker Hub, provides access to a vast collection of pre-built images.
    - Users can also create private registries for security and control.

Docker Container Life Cycle

The life cycle of a container follows these steps:

1. Create: A container is created from an image using the docker create command.
2. Start: The container starts running using the docker start command.
3. Run: A new container can be started directly using docker run.
4. Pause/Unpause: Containers can be temporarily paused and resumed.
5. Stop: The container can be stopped using docker stop.
6. Restart: A stopped container can be restarted.
7. Kill: A container can be forcefully stopped using docker kill.
8. Remove: Containers that are no longer needed can be deleted using docker rm.

Benefits of Docker

1. Portability
    - Containers can run on any platform that supports Docker.
    - Applications behave consistently across different environments.
2. Efficiency
    - Containers share the host OS kernel, reducing overhead and improving performance.

- They consume fewer resources compared to virtual machines.

3. Isolation
   - Each container runs in its own isolated environment, preventing dependency conflicts.
4. Scalability
   - Applications can be scaled up quickly by launching multiple containers.
   - Docker enables automatic load balancing in large-scale deployments.
5. Consistency
   - Ensures that the application runs the same way in development, testing, and production.
   - Eliminates the "works on my machine" problem.

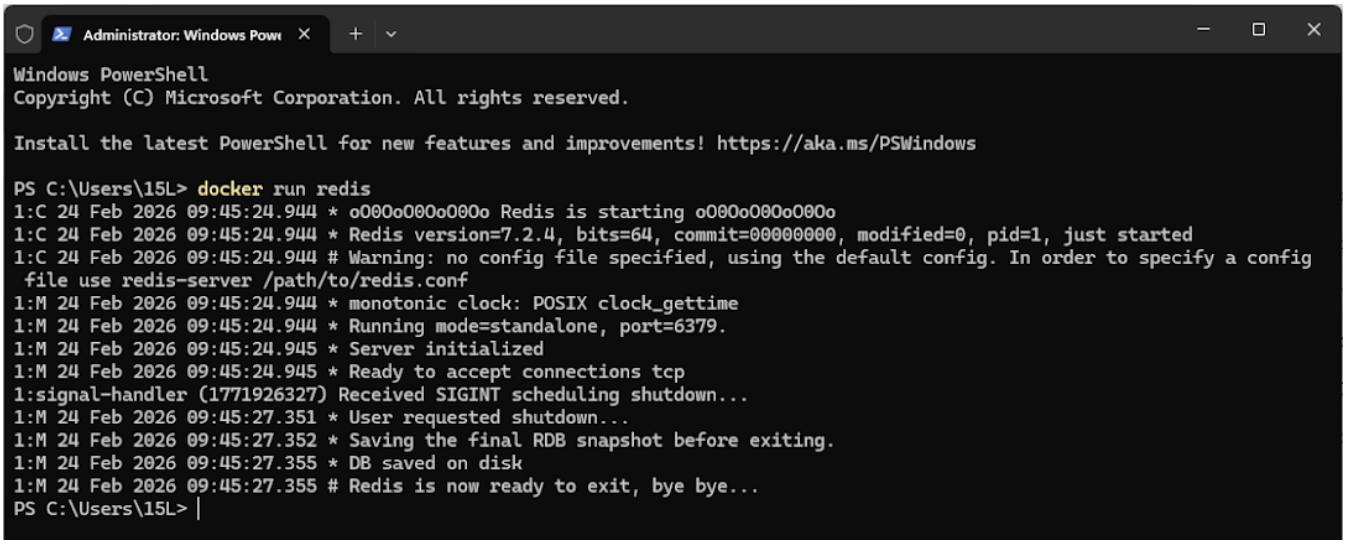
Docker Engine

At the core of Docker is the Docker Engine, which is responsible for building, running, and managing containers. It consists of the Docker daemon, which manages containers, images, networks, and volumes, and the Docker client, which allows users to interact with the daemon through the Docker API.

Docker Images

Docker images are read-only templates used to create containers. They contain the application code, runtime, libraries, dependencies, and other files needed to run the application. Images are built using Dockerfiles, which are text files that define the steps needed to create the image.

Implementation:



```
PS C:\Users\15L> docker images
REPOSITORY    TAG       IMAGE ID       CREATED         SIZE
node          latest    8bbb9d1bc633   23 months ago   1.1GB
t13-49        latest    acb00e2365d5   23 months ago   912MB
hello-world   latest    7e4af798f449   23 months ago   13.3kB
<none>        <none>    9c469d82cc74   23 months ago   13.3kB
ubuntu        latest    ca2b0f26964c   24 months ago   77.9MB
redis         latest    170a1e90f843   2 years ago     138MB
redis         <none>    d1397258b209   2 years ago     138MB
hello-world   <none>    d2c94e258dcb   2 years ago     13.3kB
PS C:\Users\15L>
PS C:\Users\15L> docker pull redis
Using default tag: latest
latest: Pulling from library/redis
0c8d55a45c0d: Pull complete
9a00fba7918f: Pull complete
253eb0a88de9: Pull complete
2ccab22a2cd8: Pull complete
5a312569f2e0: Pull complete
4f4fb700ef54: Pull complete
fb0cbf9357d9: Pull complete
Digest: sha256:dc123d5a424cbe44d681822c2a620f3f44bdba240b8a58a0eee0048a285532ed
Status: Downloaded newer image for redis:latest
docker.io/library/redis:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview redis
PS C:\Users\15L>
```

```
PS C:\Users\15L> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\15L>
```

```
PS C:\Users\15L> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS                      PORTS     NAMES
459c31b57a65   170a1e90f843   "docker-entrypoint.s…"   3 seconds ago   Exited (0) 2 minutes ago              xenodochia
l_margulis
850b23f5b27f   170a1e90f843   "docker-entrypoint.s…"   4 minutes ago   Exited (0) 5 minutes ago              gifted_sha
w
5d59addfe1bf   170a1e90f843   "docker-entrypoint.s…"   23 months ago   Exited (0) 23 months ago              lucid_jemi
son
514721e42365   170a1e90f843   "docker-entrypoint.s…"   23 months ago   Exited (0) 23 months ago              youthful_g
oodall
b6cdfdb8332b   170a1e90f843   "docker-entrypoint.s…"   23 months ago   Exited (0) 23 months ago              determined
_elion
9b50ed88f131   170a1e90f843   "docker-entrypoint.s…"   23 months ago   Exited (0) 23 months ago              beautiful_
chaum
2a945499b45e   170a1e90f843   "docker-entrypoint.s…"   23 months ago   Exited (0) 23 months ago              flamboyant
_liskov
```

```
PS C:\Users\15L> docker start 459c31b57a65
459c31b57a65
PS C:\Users\15L>
```

```
PS C:\Users\15L> docker start 459c31b57a65
459c31b57a65
PS C:\Users\15L> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS          PORTS      NAMES
459c31b57a65   170a1e90f843   "docker-entrypoint.s…"   About a minute ago   Up 29 seconds   6379/tcp   xenodochial_margu
lis
```

```
PS C:\Users\15L> docker images
REPOSITORY    TAG       IMAGE ID       CREATED         SIZE
redis         latest    f0acdeb5ced3   16 hours ago    140MB
node          latest    8bbb9d1bc633   23 months ago   1.1GB
t13-49        latest    acb00e2365d5   23 months ago   912MB
hello-world   latest    7e4af798f449   23 months ago   13.3kB
<none>        <none>    9c469d82cc74   23 months ago   13.3kB
ubuntu        latest    ca2b0f26964c   24 months ago   77.9MB
redis         <none>    170a1e90f843   2 years ago     138MB
redis         <none>    d1397258b209   2 years ago     138MB
hello-world   <none>    d2c94e258dcb   2 years ago     13.3kB
PS C:\Users\15L>
```

```
PS C:\Users\15L> docker stop 459c31b57a65
459c31b57a65
PS C:\Users\15L> docker rm 459c31b57a65
459c31b57a65
PS C:\Users\15L>
```

```
PS C:\Users\15L> docker restart 5d59addfe1bf
5d59addfe1bf
PS C:\Users\15L>
```

```
PS C:\Users\15L> docker inspect 5d59addfe1bf
[
    {
        "Id": "5d59addfe1bfa4bd83d1436db097dc935310d30077a0e491fbb4895829e69f11",
        "Created": "2024-03-26T08:44:53.058700519Z",
        "Path": "docker-entrypoint.sh",
        "Args": [
            "redis-server"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 1876,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-02-24T09:57:04.82336948Z",
            "FinishedAt": "2024-03-26T08:46:49.369075895Z"
        },
        "Image": "sha256:170a1e90f8436daa6778aeea3926e716928826c215ca23a8dfd8055f663f9428",
        "ResolvConfPath": "/var/lib/docker/containers/5d59addfe1bfa4bd83d1436db097dc935310d30077a0e491fbb4895829e69f11/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/5d59addfe1bfa4bd83d1436db097dc935310d30077a0e491fbb4895829e69f11/hostname",
        "HostsPath": "/var/lib/docker/containers/5d59addfe1bfa4bd83d1436db097dc935310d30077a0e491fbb4895829e69f11/hosts",
        "LogPath": "/var/lib/docker/containers/5d59addfe1bfa4bd83d1436db097dc935310d30077a0e491fbb4895829e69f11/5d59addfe1bfa4bd83d1436db097
dc935310d30077a0e491fbb4895829e69f11-json.log",
        "Name": "/lucid_jemison",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
```

```
PS C:\Users\15L> docker commit 5d59addfe1bf new_image_name:redis3
sha256:c133193b1b3e364104de6943d82e27a2635cd03660fb38c0bc9851e872edcce4
PS C:\Users\15L> docker images
REPOSITORY        TAG       IMAGE ID       CREATED                  SIZE
new_image_name    redis3    c133193b1b3e   Less than a second ago   138MB
redis             latest    f0acdeb5ced3   16 hours ago             140MB
node              latest    8bbb9d1bc633   23 months ago            1.1GB
t13-49            latest    acb00e2365d5   23 months ago            912MB
hello-world       latest    7e4af798f449   23 months ago            13.3kB
<none>            <none>    9c469d82cc74   23 months ago            13.3kB
ubuntu            latest    ca2b0f26964c   24 months ago            77.9MB
redis             <none>    170a1e90f843   2 years ago              138MB
redis             <none>    d1397258b209   2 years ago              138MB
hello-world       <none>    d2c94e258dcb   2 years ago              13.3kB
PS C:\Users\15L>
```

```
student@LAB08-002:/mnt/c/Users/15L$ ^C
student@LAB08-002:/mnt/c/Users/15L$ exit
logout
PS C:\Users\15L> bash
student@LAB08-002:/mnt/c/Users/15L$ mkdir myapp
student@LAB08-002:/mnt/c/Users/15L$ cd myapp
student@LAB08-002:/mnt/c/Users/15L/myapp$ echo "Hello World" > index.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ ls
index.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ cat index.html
Hello World
student@LAB08-002:/mnt/c/Users/15L/myapp$ touch demo.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ ls
demo.html  index.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ vi demo.html
student@LAB08-002:/mnt/c/Users/15L/myapp$
```

```
student@LAB08-002: /mnt/c/

Hello World! This is programmer.
~
~
~
~
~
~
~
~
~
```

```
student@LAB08-002:/mnt/c/Users/15L$ ^C
student@LAB08-002:/mnt/c/Users/15L$ exit
logout
PS C:\Users\15L> bash
student@LAB08-002:/mnt/c/Users/15L$ mkdir myapp
student@LAB08-002:/mnt/c/Users/15L$ cd myapp
student@LAB08-002:/mnt/c/Users/15L/myapp$ echo "Hello World" > index.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ ls
index.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ cat index.html
Hello World
student@LAB08-002:/mnt/c/Users/15L/myapp$ touch demo.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ ls
demo.html  index.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ vi demo.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ vi demo.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ cat demo.html

Hello World! This is programmer.

student@LAB08-002:/mnt/c/Users/15L/myapp$
student@LAB08-002:/mnt/c/Users/15L/myapp$ touch Dockerfile
student@LAB08-002:/mnt/c/Users/15L/myapp$ ls -l
total 0
-rwxrwxrwx 1 student student  0 Feb 24 15:47 Dockerfile
-rwxrwxrwx 1 student student 35 Feb 24 15:46 demo.html
-rwxrwxrwx 1 student student 12 Feb 24 15:41 index.html
student@LAB08-002:/mnt/c/Users/15L/myapp$ vi Dockerfile
student@LAB08-002:/mnt/c/Users/15L/myapp$
```

```
student@LAB08-002: /mnt/c/

FROM nginx
COPY index.html /usr/share/nginx/html
~
~
~
~
~
~
~
~
~
~
~
```

```
student@LAB08-002: /mnt/c/

student@LAB08-002:/mnt/c/Users/15L/myapp$ vi Dockerfile
student@LAB08-002:/mnt/c/Users/15L/myapp$ cat Dockerfile
FROM nginx
COPY index.html /usr/share/nginx/html
student@LAB08-002:/mnt/c/Users/15L/myapp$ docker info
Client:
 Version:    25.0.3
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.12.1-desktop.4
    Path:     /usr/local/lib/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v2.24.6-desktop.1
    Path:     /usr/local/lib/docker/cli-plugins/docker-compose
  debug: Get a shell into any image or container. (Docker Inc.)
    Version:  0.0.24
    Path:     /usr/local/lib/docker/cli-plugins/docker-debug
  dev: Docker Dev Environments (Docker Inc.)
    Version:  v0.1.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-dev
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.22
    Path:     /usr/local/lib/docker/cli-plugins/docker-extension
  feedback: Provide feedback, right in your terminal! (Docker Inc.)
    Version:  v1.0.4
    Path:     /usr/local/lib/docker/cli-plugins/docker-feedback
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.0.1
    Path:     /usr/local/lib/docker/cli-plugins/docker-init
  sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
    Version:  0.6.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-sbom
  scout: Docker Scout (Docker Inc.)
    Version:  v1.5.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-scout

Server:
 Containers: 14
  Running: 1
  Paused: 0
```

```
student@LAB08-002: /mnt/c/
ERROR: "docker buildx build" requires exactly 1 argument.
See 'docker buildx build --help'.

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Start a build
student@LAB08-002:/mnt/c/Users/15L/myapp$ docker build -t myapp .
[+] Building 5.1s (7/7) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 86B                                                                                           0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                                               2.2s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 49B                                                                                              0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208          2.7s
 => => resolve docker.io/library/nginx:latest@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208          0.0s
 => => sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208 10.23kB / 10.23kB                             0.0s
 => => sha256:514a9c2814250e61396ef4d6125ece1a8fbb3b0964a2ab441e9f7acf0b66b8b5 2.29kB / 2.29kB                               0.0s
 => => sha256:5cdef4ac3335f68428701c14c5f12992f5e3669ce8ab7309257d263eb7a856b1 9.09kB / 9.09kB                               0.0s
 => => sha256:46bf3a120c8ebc0c207a3a819bb44ad7e6bbcd0b27c2c8e7430b8baea6c321d2 33.13MB / 33.13MB                             1.9s
 => => sha256:4f4efe02d542e6a146723a61b42b2e063b2f89f58430dd4d9a8aad7ac26c905c 625B / 625B                                   0.8s
 => => sha256:7b6cb8ccac7b09e037d667024c9afc9c0429738a2b7b24c2d41f5bce008c139c 955B / 955B                                   0.6s
 => => sha256:f73400a233fdfe4b0e1ebf8a8035349bf8b8cfdc85da665c97b4099d1e8ae7df 404B / 404B                                   1.1s
 => => sha256:47cd406a84ef91b76fa7a653ed5e89663af9f8c2644a0ed0387fa0b1395e5114 1.21kB / 1.21kB                               1.3s
 => => sha256:bae5a1799a803b0a3df8179c56dd95fdb6f88c7aa10600d101196c723994f679 1.40kB / 1.40kB                               1.7s
 => => extracting sha256:46bf3a120c8ebc0c207a3a819bb44ad7e6bbcd0b27c2c8e7430b8baea6c321d2                                     0.5s
 => => extracting sha256:4f4efe02d542e6a146723a61b42b2e063b2f89f58430dd4d9a8aad7ac26c905c                                     0.0s
 => => extracting sha256:7b6cb8ccac7b09e037d667024c9afc9c0429738a2b7b24c2d41f5bce008c139c                                     0.0s
 => => extracting sha256:f73400a233fdfe4b0e1ebf8a8035349bf8b8cfdc85da665c97b4099d1e8ae7df                                     0.0s
 => => extracting sha256:47cd406a84ef91b76fa7a653ed5e89663af9f8c2644a0ed0387fa0b1395e5114                                     0.0s
 => => extracting sha256:bae5a1799a803b0a3df8179c56dd95fdb6f88c7aa10600d101196c723994f679                                     0.0s
 => [2/2] COPY index.html /usr/share/nginx/html                                                                               0.1s
 => exporting to image                                                                                                        0.0s
 => => exporting layers                                                                                                       0.0s
 => => writing image sha256:a5a22179b024370e5684307ba366c66ae4c69e6e9cf900dcba744f2d613b4e8d                                  0.0s
 => => naming to docker.io/library/myapp                                                                                      0.0s

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview
student@LAB08-002:/mnt/c/Users/15L/myapp$
```

```
student@LAB08-002:/mnt/c/Users/15L/myapp$ docker images
REPOSITORY        TAG       IMAGE ID       CREATED              SIZE
myapp             latest    a5a22179b024   About a minute ago   161MB
new_image_name    redis3    c133193b1b3e   25 minutes ago       138MB
redis             latest    f0acdeb5ced3   17 hours ago         140MB
node              latest    8bbb9d1bc633   23 months ago        1.1GB
t13-49            latest    acb00e2365d5   23 months ago        912MB
hello-world       latest    7e4af798f449   23 months ago        13.3kB
<none>            <none>    9c469d82cc74   23 months ago        13.3kB
ubuntu            latest    ca2b0f26964c   24 months ago        77.9MB
redis             <none>    d1397258b209   2 years ago          138MB
redis             <none>    170a1e90f843   2 years ago          138MB
hello-world       <none>    d2c94e258dcb   2 years ago          13.3kB
student@LAB08-002:/mnt/c/Users/15L/myapp$
```

```
student@LAB08-002:/mnt/c/Users/15L/myapp$ docker run -p 8081:80 myapp
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/02/24 10:29:05 [notice] 1#1: using the "epoll" event method
2026/02/24 10:29:05 [notice] 1#1: nginx/1.29.5
2026/02/24 10:29:05 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/02/24 10:29:05 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/02/24 10:29:05 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/02/24 10:29:05 [notice] 1#1: start worker processes
2026/02/24 10:29:05 [notice] 1#1: start worker process 29
2026/02/24 10:29:05 [notice] 1#1: start worker process 30
2026/02/24 10:29:05 [notice] 1#1: start worker process 31
2026/02/24 10:29:05 [notice] 1#1: start worker process 32
2026/02/24 10:29:05 [notice] 1#1: start worker process 33
2026/02/24 10:29:05 [notice] 1#1: start worker process 34
2026/02/24 10:29:05 [notice] 1#1: start worker process 35
2026/02/24 10:29:05 [notice] 1#1: start worker process 36
2026/02/24 10:29:05 [notice] 1#1: start worker process 37
2026/02/24 10:29:05 [notice] 1#1: start worker process 38
2026/02/24 10:29:05 [notice] 1#1: start worker process 39
2026/02/24 10:29:05 [notice] 1#1: start worker process 40
2026/02/24 10:29:05 [notice] 1#1: start worker process 41
2026/02/24 10:29:05 [notice] 1#1: start worker process 42
2026/02/24 10:29:05 [notice] 1#1: start worker process 43
2026/02/24 10:29:05 [notice] 1#1: start worker process 44
2026/02/24 10:29:05 [notice] 1#1: start worker process 45
2026/02/24 10:29:05 [notice] 1#1: start worker process 46
2026/02/24 10:29:05 [notice] 1#1: start worker process 47
2026/02/24 10:29:05 [notice] 1#1: start worker process 48
```

```
student@LAB08-002: /mnt/c/
2026/02/24 10:29:05 [notice] 1#1: start worker process 46
2026/02/24 10:29:05 [notice] 1#1: start worker process 47
2026/02/24 10:29:05 [notice] 1#1: start worker process 48
^C2026/02/24 10:31:11 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/02/24 10:31:11 [notice] 29#29: exiting
2026/02/24 10:31:11 [notice] 32#32: exiting
2026/02/24 10:31:11 [notice] 33#33: exiting
2026/02/24 10:31:11 [notice] 35#35: exiting
2026/02/24 10:31:11 [notice] 31#31: exiting
2026/02/24 10:31:11 [notice] 32#32: exit
2026/02/24 10:31:11 [notice] 29#29: exit
2026/02/24 10:31:11 [notice] 33#33: exit
2026/02/24 10:31:11 [notice] 37#37: exiting
2026/02/24 10:31:11 [notice] 31#31: exit
2026/02/24 10:31:11 [notice] 35#35: exit
2026/02/24 10:31:11 [notice] 30#30: exiting
2026/02/24 10:31:11 [notice] 38#38: exiting
2026/02/24 10:31:11 [notice] 39#39: exiting
2026/02/24 10:31:11 [notice] 34#34: exiting
2026/02/24 10:31:11 [notice] 36#36: exiting
2026/02/24 10:31:11 [notice] 42#42: exiting
2026/02/24 10:31:11 [notice] 38#38: exit
2026/02/24 10:31:11 [notice] 41#41: exiting
2026/02/24 10:31:11 [notice] 39#39: exit
2026/02/24 10:31:11 [notice] 43#43: exiting
2026/02/24 10:31:11 [notice] 44#44: exiting
2026/02/24 10:31:11 [notice] 30#30: exit
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2026/02/24 10:31:11 [notice] 40#40: exiting
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2026/02/24 10:31:11 [notice] 41#41: exit
2026/02/24 10:31:11 [notice] 46#46: exiting
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2026/02/24 10:31:11 [notice] 44#44: exit
2026/02/24 10:31:11 [notice] 43#43: exit
2026/02/24 10:31:11 [notice] 48#48: exiting
2026/02/24 10:31:11 [notice] 40#40: exit
2026/02/24 10:31:11 [notice] 46#46: exit
2026/02/24 10:31:11 [notice] 48#48: exit
2026/02/24 10:31:11 [notice] 47#47: exiting
2026/02/24 10:31:11 [notice] 47#47: exit
student@LAB08-002:/mnt/c/Users/15L/myapp$ docker run -d -p 8081:80 myapp
a3ef951fe5fe4291881bf414757307bcca996d293e2de73f9093a7447cbec170
student@LAB08-002:/mnt/c/Users/15L/myapp$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                   NAMES
a3ef951fe5fe   myapp          "/docker-entrypoint.…"   28 seconds ago   Up 27 seconds   0.0.0.0:8081->80/tcp    happy_shockley
5d59addfe1bf   170a1e90f843   "docker-entrypoint.s…"   23 months ago    Up 35 minutes   6379/tcp                lucid_jemison
student@LAB08-002:/mnt/c/Users/15L/myapp$
```

Conclusion:

Hence, we have successfully studied Docker Architecture and Container Life Cycle, installed Docker and executed docker commands to manage images and interact with containers.

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